

```

In[*]:= dataV =
  {35, 44, 51, 61, 76, 90, 103, 110, 124, 135, 147, 159, 172, 188, 200, 211, 227, 238, 248};

In[*]:= dataVd = {0.1, 0.4, 0.8, 1.9, 4.8, 10.4, 17.4, 23.1, 36.3,
  50.4, 67.6, 89.4, 117.6, 157.4, 193.6, 228, 285, 319, 343} * 10^(-3);

In[*]:= dataI = {240, 274, 296, 326, 366, 404, 435, 451,
  482, 506, 529, 552, 577, 606, 627, 646, 673, 691, 707} * 10^(-3);

In[*]:= (*PARTE 4.1*)

In[*]:= dataLnP = Log[dataI * dataV];
           |logaritmo

In[*]:= dataLnR = Log[dataV / dataI];
           |logaritmo

In[*]:= datap1 = Transpose[{dataLnR, dataLnP}];
           |transposición

In[*]:= fit1 = NonlinearModelFit[datap1, A + 4 * G * x, {A, G}, x]
           |ajusta a modelo no lineal

Out[*]=
  FittedModel[-15.+3.45 x]

In[*]:= fit1[{"BestFit", "ParameterTable"}]

Out[*]=
  { -15.0493 + 3.44862 x, 

|   | Estimate | Standard Error | t-Statistic | P-Value                     |
|---|----------|----------------|-------------|-----------------------------|
| A | -15.0493 | 0.0462538      | -325.363    | 1.06741 × 10 <sup>-33</sup> |
| G | 0.862156 | 0.00208835     | 412.84      | 1.86427 × 10 <sup>-35</sup> |

 }

In[*]:= plotfit1 =
  Plot[fit1[x], {x, 4.4, 5.9}, BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
  |representación gráfica |estilo base |familia de tipo de letra |tamaño de tipo de letra
  PlotStyle → {Blend[{Blue, White}], Thick},
  |estilo de repre... |mezcla... |azul |blanco |grueso
  PlotRange → {{4.9, 5.9}, {2, 5.6}}, Frame → True,
  |rango de representación |marco |verdadero
  FrameLabel → {Style["Ln( R / Ω )", Bold, 16], Style["Ln( P / W )", 16, Bold]}};
  |etiqueta de marco |estilo |negrita |estilo |negrita

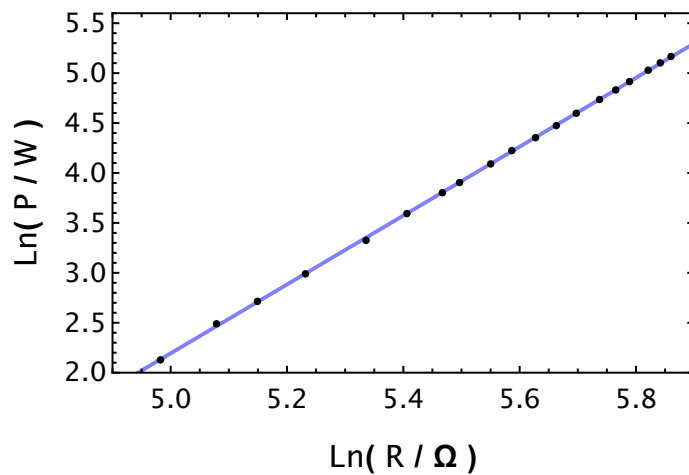
In[*]:= plotdata1 = ListPlot[datap1, PlotStyle → {Black, Thickness[0.003] }];
  |representación de lista |estilo de repre... |negro |grosor

```

```
In[*]:= Show[plotfit1, plotdata1]
```

[muestra](#)

```
Out[*]=
```



```
In[*]:= S[A_, G_, R0_, T0_, sigm_] := Exp[A] * R0^(4 * G) / (sigm * T0^4)
```

[exponencial](#)

```
In[*]:= S[-15.049257050057296, 0.862156, 25, 295, 5.67 * 10^-8] (*valor de S*)
```

```
Out[*]=
```

0.0000449044

```
In[*]:= D[S[A, G, R0, T0, sigm], A]
```

[deriva](#)

```
Out[*]=
```

$$\frac{e^A R0^{4G}}{\text{sigm } T0^4}$$

```
In[*]:= dA[A_, G_, R0_, T0_, sigm_] := \frac{e^A R0^{4G}}{\text{sigm } T0^4}
```

```
In[*]:= dAA = dA[-15.049257050057296, 0.862156, 25, 295, 5.67 * 10^-8]
```

```
Out[*]=
```

0.0000449044

```
In[*]:= D[S[A, G, R0, T0, sigm], G]
```

[deriva](#)

```
Out[*]=
```

$$\frac{4 e^A R0^{4G} \text{Log}[R0]}{\text{sigm } T0^4}$$

```
In[*]:= dG[A_, G_, R0_, T0_, sigm_] := \frac{4 e^A R0^{4G} \text{Log}[R0]}{\text{sigm } T0^4}
```

```
In[*]:= dGG = dG[-15.049257050057296, 0.862156, 25, 295, 5.67 * 10^-8]
```

```
Out[*]=
```

0.000578167

```
In[*]:= D[S[A, G, R0, T0, sigm], R0]
```

[\[deriva\]](#)

```
Out[*]=
```

$$\frac{4 e^A G R_0^{-1+4 G}}{\text{sigm } T_0^4}$$

```
In[*]:= dR0[A_, G_, R0_, T0_, sigm_] := \frac{4 e^A G R_0^{-1+4 G}}{\text{sigm } T_0^4}
```

```
In[*]:= dRR = dR0[-15.049257050057296, 0.862156, 25, 295, 5.67 * 10^-8]
```

```
Out[*]=
```

$$6.19434 \times 10^{-6}$$

```
In[*]:= errSd1 = Sqrt[(0.10217155108411698 * dAA)^2 + (0.00125 * dGG)^2 + (dRR)^2] (*error de S*)
```

[\[raíz cuadrada\]](#)

```
Out[*]=
```

$$7.74219 \times 10^{-6}$$

```
In[*]:= (*PARTE 4.2*)
```

```
In[*]:= datap2 = Transpose[{(dataV / dataI)^(-0.8621819475302489), Log[dataVd]}];
```

[\[transposición\]](#)

[\[logaritmo\]](#)

```
In[*]:= fit2 = NonlinearModelFit[datap2, A - B * ((25^0.8621819) / 295) * x, {A, B}, x]
```

[\[ajusta a modelo no lineal\]](#)

```
Out[*]=
```

FittedModel[6.13 - 1130. x]

```
In[*]:= fit2[{"BestFit", "ParameterTable"}]
```

```
Out[*]=
```

		Estimate	Standard Error	t-Statistic	P-Value
{6.12558 - 1125.59 x,	A	6.12558	0.065501	93.5188	1.68683 × 10 ⁻²⁴
	B	20697.7	134.227	154.199	3.46151 × 10 ⁻²⁸

```
In[*]:= h = 20697.704986641365 * (578 * 10^(-9) * 1.380658 * 10^(-23)) / (299792458)
```

```
Out[*]=
```

$$5.50954 \times 10^{-34}$$

```
In[*]:= errh = 134.22729915034725 * (578 * 10^(-9) * 1.380658 * 10^(-23)) / (299792458)
```

```
Out[*]=
```

$$3.57301 \times 10^{-36}$$

```
In[*]:= difrel100 = (6.62607 - h * 10^34) / (6.62607) * 100
```

```
Out[*]=
```

$$16.8505$$

```

In[*]:= plotfit2 = Plot[fit2[x], {x, 0.003, 0.026},
  representación gráfica
  BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
  estilo base familia de tipo de letra tamaño de tipo de letra
  PlotStyle → {Blend[{Blue, White}], Thick},
  estilo de repre... mezcla... azul blanco grueso
  Frame → True, PlotRange → {{0.006, 0.014}, {-10, 0}},
  marco verd... rango de representación
  FrameLabel → {Style["( R / Ω ) ^ (-γ)", Bold, 16], Style["Ln( Vd / V )", 16, Bold]}}];
  etiqueta de marco estilo negrita estilo negrita

```

```

In[*]:= plotdata2 = ListPlot[datap2, PlotStyle → {Black, Thickness[0.003] }];
  representación de lista estilo de repre... negro grosor

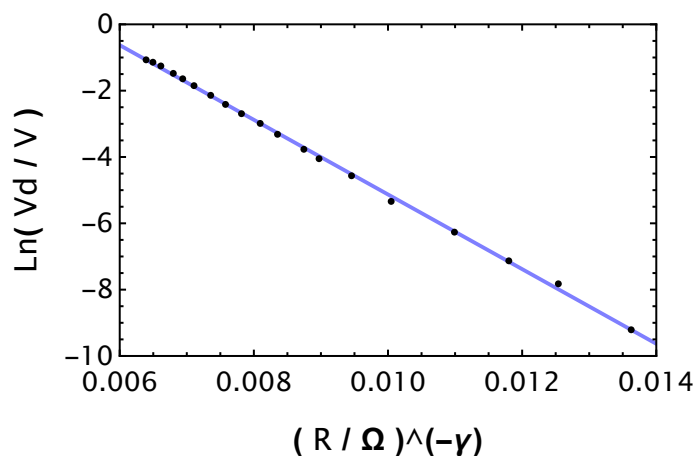
```

```

In[*]:= Show[plotfit2, plotdata2]
  muestra

```

Out[*]=



```

In[*]:= (*PARTE 5*)

```

```

In[*]:= (*hc/KTLambda mucho mayor a 1?*)

```

```

In[*]:= 20697.704986641365 * ((25^0.8621819) / 295) * ((dataV / dataI) ^ (-0.8621819475302489))

```

Out[*]=

```

{15.337, 14.1144, 13.2831, 12.371, 11.3088, 10.6438, 10.0985, 9.84383, 9.4015, 9.11103,
 8.79681, 8.52861, 8.28012, 8.00005, 7.81052, 7.65262, 7.44344, 7.31034, 7.19611}

```

```

In[*]:= (*Rango de Ts*)

```

```

In[*]:= 295 * ((dataV / dataI) / 25) ^ 0.8621819

```

Out[*]=

```

{1349.53, 1466.43, 1558.2, 1673.08, 1830.23, 1944.58, 2049.57, 2102.61, 2201.53, 2271.72,
 2352.86, 2426.85, 2499.69, 2587.2, 2649.98, 2704.65, 2780.66, 2831.29, 2876.23}

```

```

In[*]:= (*suponemos que L=15mm y diam=1, y que es un cilindro*)

```

```

In[*]:= Steo = N[2 * Pi * (1 / 2) * 15] (*en mm^2*)
  valo... número pi

```

Out[*]=

```

47.1239

```

```
In[*]:= errSteo = 2 * Pi * 0.5 * 1
```

↳ número pi

```
Out[*]=
```

3.14159